

John Peter D. Pestaño

 johnpeter.diongzon.pestano@gmail.com  (+63) 9606728563  john-peter-pestano

 FloatingDust36  Portfolio

Technical Skills

Programming

- Python, TypeScript/JavaScript, C/C++, C#, Dart, Bash

Frontend & Mobile

- React.js, React Native (Expo), Flutter, Tailwind CSS

Backend & AI

- Node.js (Express), FastAPI, REST APIs, TensorFlow, PyTorch, OpenCV

Databases

- PostgreSQL, Supabase, Firebase, Neo4j, SQLAlchemy

Hardware

- Arduino, ESP32, Verilog

DevOps & Tools

- Git/GitHub, Docker, GitHub Actions (CI/CD), Postman, Figma, Cisco Packet Tracer

Education

Bachelor of Science in Computer Engineering

Cebu Institute of Technology-University 

2023 – Expected May 2027 | Cebu City, Philippines

Certifications and Awards

Associate Python Developer

DataCamp

29/03/2026

1st Runner Up, Competitive Programming (Python)

CpE Regional Challenge

2026

AutoMATHic

University of the Philippines Civil Engineering Society (UPCES)

2026

Top 10 Finalist

MCIA CEB-i Hacks Hackathon

2026

Shortlisted Finalist (Top 28)

IBPAP: Can You HackIT Hackathon

2024

DOST-SEI RA 7687 Scholar

2023

Projects

Filipino Sign Language (FSL) AI Classifier

Machine Learning Engineer

04/2026 – 05/2026

- Engineered an end-to-end machine learning pipeline using Python, OpenCV, and MediaPipe to automate the extraction of 63 spatial 3D hand landmark coordinates per frame, collecting a 9,600-sample custom training dataset (400 samples × 24 FSL letters).
- Trained a deep neural network using TensorFlow and Keras with early stopping and model checkpointing to classify 24 distinct alphabetical FSL hand gestures in real-time.
- Deployed the trained classification model to a custom web interface using TensorFlow.js, re-implementing the Python normalization pipeline in JavaScript to enable zero-latency, privacy-preserving sign language translation directly within the browser without requiring server-side API calls.

ReliefMatch AI

Full-Stack Developer

03/2026 – 05/2026

- Engineered a full-stack disaster relief management platform using React, FastAPI, and PostgreSQL to streamline supply allocation and logistics for Philippine Local Government Units, supporting three distinct user roles with JWT-based authentication.
- Developed an XGBoost regression model trained on 2,000+ synthetic data points across seven disaster-related features to prioritize aid distribution, utilizing SHAP (SHapley Additive exPlanations) to generate transparent, plain-English justifications for each AI decision.
- Implemented a logistics optimization engine using Google OR-Tools to solve Capacitated Vehicle Routing Problems (CVRP), computing the most efficient multi-truck delivery sequences based on Haversine distance formulas.

Tuklascope Mobile

Mobile & Lead Backend Developer

03/2026 – 05/2026

- Engineered a cross-platform educational mobile application using Flutter/Riverpod and Python FastAPI, integrating Supabase for JWT-based authentication and Row Level Security-aware PostgreSQL data management.
- Architected an AI content generation pipeline using LangChain and Google Gemini 2.5 Flash for multimodal object recognition, structured output generation, and context-aware conversational tutoring with full conversation history management.
- Implemented a Neo4j AuraDB graph database with async connection pooling to autonomously map hierarchical Strand → Domain → Skill learning pathways, powering a real-time RPG-style XP progression system and an interactive force-directed skill tree visualization.